



Research paper

A multi-attribute quantum group consensus model considering psychological preference

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ABSTRACT

In response to the issues of information uncertainty and opinion interference in meteorological disaster emergency rescue, this paper proposes a multi-attribute quantum group decision-making model. Firstly, the paper employs the linguistic Z-numbers to represent the evaluation information of decision-makers in complex uncertain decision-making environments. Secondly, a group consensus model based on minimal adjustments is constructed, considering the psychological state and adjustment willingness of decision-makers when facing group preference pressure. On this basis, the weights of decision-makers are determined by incorporating their marginal contribution levels. Meanwhile, an optimization model is developed based on the marginal contributions from Shapley to determine the optimal attributes' weights. Furthermore, a quantum-like Bayesian network is constructed to compute the quantum probabilities of alternatives and rank them, while utilizing Deng entropy measure to address the interference terms in the quantum probabilities. Finally, through the analysis of simulation data, the applicability of the model in evaluating emergency alternatives for meteorological disasters is verified, and its stability and practicality are further demonstrated through sensitivity analysis, comparative analysis and simulation analysis. The result shows that combining quantum computing with artificial intelligence achieves efficient consensus in intelligent group decision-making processes.

1. Introduction

In the context of highly uncertain and complex disasters, such as meteorological disasters, group decision-making becomes crucial (Yucheng Dong, 2016; Meng et al., 2024; Teng and Liu, 2021). Meteorological disasters like including hurricanes, floods, and droughts often pose significant threats to life and property. Due to their sudden onset and wide-ranging impact, they command considerable public attention. Group decision-making (GDM) harnesses collective wisdom and resources, enabling more effective responses to disaster risks and emergency situations. However, the knowledge limitations and subjective biases of decision makers (DMs) may affect the quality of decision making, and obtaining accurate quantitative data is often challenging. Additionally, obtaining accurate quantitative data in such scenarios is often challenging, further complicating the decision-making process.

As a result, addressing these uncertainties becomes critical. Linguistic information has emerged as an essential tool for managing

such uncertainty (Delgado et al., 1992). It allows decision makers to express their judgments more flexibly, even when precise numerical data is unavailable. Linguistic information can be represented by binary linguistic numbers (Geng et al., 2017), hesitant fuzzy linguistic term sets (Wu et al., 2019) and probabilistic linguistic term sets (Zhang et al., 2024b), which provide a way to capture and process the inherent vagueness in disaster-related decision-making. However, when dealing with uncertainty, these linguistic terms can only express the evaluation value, and it is difficult to reflect the reliability of information at the same time. By contrast, the linguistic Z-numbers (LZNs) can not only express uncertainty, but also more accurately describe the reliability of information by introducing both evaluation and reliability (Chai et al., 2023; Zhang and Li, 2023b; Li et al., 2020, 2023). For example, in emergency rescue decision making, which is usually faced with high uncertainty and fast decision requirements, the LZNs shows significant advantages in this case. The LZNs can comprehensively reflect the opinion of DMs through the combination of linguistic terms and confidence measures, and its linguistic variables can directly reflect the subjective

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evaluation of experts on events or programs, while confidence measures describe the degree of trust experts have in their own judgments. This dual structure is especially suitable for information ambiguity and reliability fluctuation in emergency scenarios. Compared with interval-valued fuzzy sets (Karaaslan and Karamaz, 2024) and hesitant fuzzy numbers (Wan et al., 2024), LZNs have higher representation power and applicability. In terms of information representation dimension, interval valued fuzzy sets mainly describe the uncertainty of evaluation through fixed range, while hesitancy fuzzy numbers represent the hesitancy of experts through multiple possible values. However, neither of these approaches involves modeling confidence levels and therefore has limitations in dealing with information reliability and trustworthiness. Therefore, the LZNs has stronger collaborative potential in group decision making, and its confidence dimension makes the integration process of different expert opinions more scientific and transparent, especially in emergency decision making scenarios with complex interaction of multi-source information (Ren et al., 2020; Chai et al., 2021).

In group decision-making, achieving effective consensus is crucial. To achieve this, the decision-making process needs to integrate a structured consensus process. The process usually includes two main stages: measurement and quantification of consensus level, and preference adjustment based on consensus rules (Chiclana et al., 2013; Ma et al., 2024; Meng et al., 2023). Liu and Zhu (2022) consider group consensus reaching based on the normal cloud model. Decision makers make adjustments according to the compromise interval and feedback information, but still retain their personal preferences and make adjustments within the personal cognitive interval. Scholars have introduced confidence degree one after another, in which each decision maker's assessment cognition under a certain attribute has the confidence limit of individual preference (Zha et al., 2021), and proposed a framework based on bilateral negotiation mechanism to improve group consensus through the interaction between individual decision makers, while taking into account the limited compromise behavior of decision makers in the process of reaching consensus (Cao et al., 2024). However, the above studies fail to fully consider the influence between individual preferences and group preferences that decision makers face when making adjustments after receiving feedback information. This psychological state may affect the decision-making level and the final result of the decision maker.

In multi-attribute group decision making (MAGDM), the determination of attributes' weights is a key problem. Traditionally, methods usually include expert scoring, analytic hierarchy process (AHP), entropy method, Delphi method and other methods (Erdoğan et al., 2024; Yang et al., 2023; Zhou et al., 2023). These methods determine the importance of individual attributes through subjective judgment by decision makers or objective analysis based on data. For example, AHP uses consistency judgment to determine the relative importance of attributes by constructing pairwise comparison matrix; The entropy rule assigns weights according to the dispersion of attribute values. However, these traditional methods often fail to consider the marginal contribution of each attribute to the overall decision utility. In decision-making scenarios, the influence of an attribute is not isolated; its contribution should be evaluated not only on its individual importance but also on its incremental value within different subsets of attributes. Ignoring these marginal contributions can result in unfair weight allocation, where attributes with significant incremental value are undervalued, leading to biased decisions and suboptimal outcomes.

The group interference effect refers to the opinion deviation that arises from the interaction and influence among decision makers during the group decision-making process (Gao et al., 2022; Wang et al., 2024b; Busemeyer, 2012). Traditional group decision models often overlook this effect, assuming that each decision maker evaluates information independently. In reality, however, the opinions and choices of decision makers are substantially shaped by the influence and emotions

of others. By introducing the concept of quantum interference, quantum probability theory can simulate these interactions and provide a more realistic depiction of group decision-making behavior. In modeling the dynamics of group decision making — especially when accounting for the complex psychological and social interactions among decision makers — quantum models can highlight paradoxes, sub-optimal choices, and irrational behaviors in human decision making (Wu et al., 2021b; Jiang and Liu, 2022; Moreira et al., 2020). Tan et al. (2020) explored group conflict and ambiguity by incorporating interference effects among decision makers, aiming to better understand how group consensus preferences are reached. The quantum interference effect will play a significant role in the calculation of the final probability, thus affecting the decision result. In quantum theory, the interference value is usually expressed by the phase angle, but this angle is often uncertain (Yan et al., 2024). In order to measure the interference angle more accurately and reflect the interaction between decision makers, She et al. (2021) introduced confidence entropy as an index to calculate probability uncertainty.

In summary, although existing studies have addressed the application scope and effectiveness of multi-attribute group decision-making, several challenges remain unresolved: (1) Most existing multi-attribute group decision-making models overlook the mutual interference effect among decision makers. In complex group decision-making scenarios, the opinions of individual decision makers often influence one another, leading to interference in the decision-making process. (2) Many group consensus models fail to comprehensively account for the psychological state of decision makers and their level of contribution to the group decision-making process. This oversight can impair the performance of consensus models, particularly when individual differences among decision makers are significant. (3) In multi-attribute decision-making problems involving uncertain information, rational weight determination is crucial. However, under complex uncertain environments, existing methods often struggle to balance adaptability and accuracy, resulting in weight allocation outcomes that lack robustness.

On this basis, a quantum group consensus decision model based on linguistic Z-numbers is proposed to enhance the accuracy and reliability of decision-making. Firstly, during the process of group opinion adjustment, the pressure exerted by group preferences on DMs is considered, and a feedback suggestion matrix is determined by integrating the psychological preferences of decision makers, thereby fully reflecting their comprehensive decision-making situation. Based on this, the decision makers' weights are calculated according to their contribution levels. Secondly, a model considering the interdependence and synergistic effect between attributes is constructed to solve the attributes' weights, which ensures the objectivity and scientificity of weight distribution. Finally, the quantum probability values of alternatives are computed and ranked the alternatives, where, the interference value measured using Deng entropy. The model effectively addresses the entanglement effect in group decision-making, allowing for a more accurate representation of complex real-world decision processes. A comparison between the proposed and existing decision models is presented in Table 1.

The main contributions of this paper are as follows:

(1) In this paper, a quantum group decision model focuses on the potential interference effects among decision makers during the group decision-making process. By introducing quantum probability, the model effectively captures the interactions between decision makers, enhancing its capacity to reflect decision interference in complex group decision-making scenarios. This makes the model's application in dynamic environments, such as meteorological disasters, more realistic and practical.

(2) In the process of forming group consensus, the proposed model incorporates the psychological states of decision makers and considers their willingness to adjust their evaluation information. Additionally, the model accounts for the contribution levels of decision makers, ensuring fairness and rationality in the decision-making process by assigning greater influence to decision makers with higher contributions.

Table 1

The comparison between the proposed method and existing decision models.

Models	Information	Decision makers' weights	Attributes' weights	Psychological preference	Group consensus	Interference effect
Wu and Liao (2024)	CN	Unknown	Unknown	×	×	×
Liao et al. (2024)	HFPR	Unknown	Unknown	×	✓	×
Zha et al. (2021)	CN	Known	Known	×	✓	×
Chang et al. (2024)	LTs	Unknown	Unknown	×	×	×
Zhang et al. (2024a)	CN	Unknown	Unknown	✓	✓	×
Wu et al. (2021a)	LDA	Known	Known	×	×	✓
Xiao et al. (2024)	LDA	Unknown	Unknown	×	✓	✓
The proposed model	LZN	Unknown	Unknown	✓	✓	✓

Note: CN is Crisp number, HFPR is Hesitant Fuzzy Preference Relation, LTs is Linguistic term sets, LDA is Linguistic distribution assessment.

(3) An optimization model based on Shapley is constructed to solve the weight of attributes, which ensures the fairness and rationality of weight distribution in multi-attribute group decision making. By accurately calculating the marginal contribution of each attribute in the overall decision, Shapley value avoids the weight bias that may be brought by the traditional weighted summation method, and makes the weight allocation of attributes more just and scientific.

(4) This paper proposes a comprehensive calculation method combining Shapley value and Minkowski distance to determine decision maker weight. In this approach, the Shapley value is used to fairly measure the historical contribution of the decision maker, while the Minkowski distance is used to flexibly measure the proximity between the decision maker and group preferences by adjusting the parameters. Through this method, the contribution difference of decision makers in multi-attribute decision-making can be accurately captured, and the sensitivity can be adjusted according to different decision scenarios (such as absolute difference, square variance, etc.), making the weight calculation more reasonable and flexible.

The main structure of this paper is as follows: Section 2 introduces the basic concepts of linguistic Z-numbers and quantum probability theory, which provides theoretical basis for the subsequent model development. In Section 3, this paper proposes a multi-attribute quantum group consensus model that combines the contribution levels of DMs and their psychological preferences. In Section 4, the validity and robustness of the model are verified by sensitivity analysis, comparative analysis and simulation analysis, taking meteorological disaster as an case. Finally, Section 5 summarizes the contributions of this paper and discusses potential directions for future research.

2. Preliminaries

2.1. Linguistic Z-numbers (LZNs)

In 2011, Zadeh (2011) proposed the concept of Z-numbers to improve uncertain, inaccurate or incomplete information representations.

Definition 1 (Wang et al., 2024a). Let X be the domain of discourse, and $A = \{A_0, A_1, \dots, A_{2g}\}$ and $B = \{B_0, B_1, \dots, B_{2g'}\}$ be two finite and fully ordered sets of discrete fuzzy linguistic terms, where g and g' are non-negative integers. In addition, let $A_{\varphi(x)} \in A$ and $B_{\Phi(x)} \in B$. The set of linguistic Z-numbers in X has the following form:

$$Z = \{(x, A_{\varphi(x)}, B_{\Phi(x)}) | x \in X\}. \quad (1)$$

Where Z is represented by the form of a binary group, $A_{\varphi(x)}$ shows the ambiguity or semantic description of an event, and $B_{\Phi(x)}$ is the confidence of $A_{\varphi(x)}$, describing the reliability assessment of the ambiguous event. And the linguistic term sets A and B are different and represent different preference information respectively.

Definition 2 (Chai et al., 2023). The distance measure of two linguistic Z-numbers $z_x = (A_{\varphi(x)}, B_{\Phi(x)})$ and $z_{x'} = (A_{\varphi(x')}, B_{\Phi(x')})$ can be defined

as:

$$d(z_x, z_{x'}) = \sqrt{\rho_1(f^*(A_{\varphi(x)}) - f^*(A_{\varphi(x')}))^2 + \rho_2(g^*(B_{\Phi(x)}) - g^*(B_{\Phi(x')}))^2}. \quad (2)$$

Where $\rho_1 \in [0, 1]$ and $\rho_2 \in [0, 1]$ represent the weight coefficients, and $\rho_1 + \rho_2 = 1$.

2.2. Quantum probability theory

The application of quantum probability theory to decision theory is a relatively new area of research, and unlike traditional decision theory, which assumes that an individual will evaluate all possibilities when faced with various options and make an optimal decision accordingly, quantum probability theory introduces concepts from quantum mechanics, such as superposition states and quantum entanglement. To explain some of the non-classical behaviors observed in human decision-making (Moreira et al., 2020), as shown in Fig. 1.

According to the total probability principle in classical probability theory, can be calculated according to the total probability theorem:

$$P(A) = P(G)P(A|G) + P(B)P(A|B). \quad (3)$$

In quantum-like Bayesian networks, probability amplitude is used to represent probability, and the magnitude of the probability is equal to the square of the projected length (Yan et al., 2024). According to Feynman's first rule, the probability amplitude of a path is the product of the probability amplitudes along that path. Thus, the probability amplitude of choosing W along path $P \rightarrow G \rightarrow W$ is $\psi_G \psi_{W|G}$, and the probability amplitude of choosing A along path $P \rightarrow B \rightarrow A$ is $\psi_B \psi_{A|B}$. In real-life scenarios, the specific path followed by a decision cannot be clearly observed, and the system remains in a superposition state along the path. According to Feynman's second rule, if no particular path of state transition is observed, the probability amplitudes of all possible paths are first summed and then squared to calculate the probability of the final transition:

$$\begin{aligned} P(A) &= |\langle A|P \rangle|^2 = |\langle G|P \rangle \cdot \langle A|G \rangle + \langle B|P \rangle \cdot \langle A|B \rangle|^2 \\ &= \left(\sqrt{P(G)P(A|G)} \cdot e^{i\alpha} + \sqrt{P(B)P(A|B)} \cdot e^{i\beta} \right) \cdot \\ &\quad \left(\sqrt{P(G)P(A|G)} \cdot e^{-i\alpha} + \sqrt{P(B)P(A|B)} \cdot e^{-i\beta} \right) \\ &= P(G)P(A|G) + P(B)P(A|B) + \\ &\quad 2\sqrt{P(G)P(A|G)P(B)P(A|B)} \cdot \cos(\alpha - \beta). \end{aligned} \quad (4)$$

3. Multi-attribute group consensus model considering quantum interference

The main goal of a multi-attribute group decision problem is to derive a consensus-based solution from multiple alternatives with multiple attributes. The typical elements of multi-attribute group decision problems are as follows.

(1) The decision maker set $D = \{d_1, d_2, \dots, d_s\} (s \geq 2)$ contains s decision makers, the weight set of the decision is $\lambda = \{\lambda_1, \lambda_2, \dots, \lambda_s\}$,

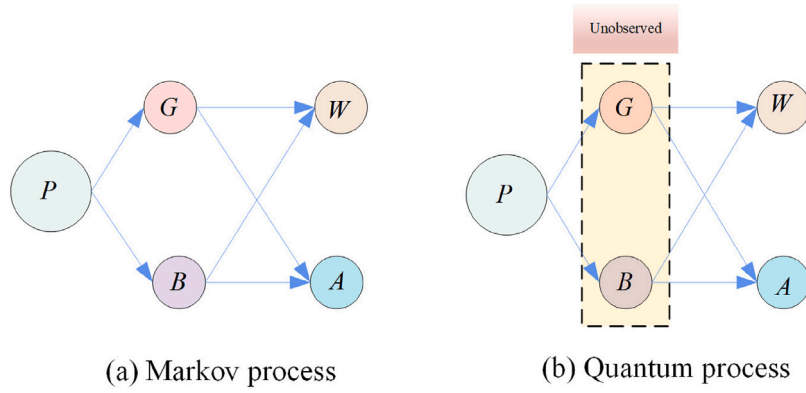


Fig. 1. Markov link path.

which satisfies the conditions $\lambda_k \geq 0 (k = 1, 2, \dots, s)$ and $\sum_{k=1}^s \lambda_k = 1$.

(2) The limited set of alternative is $A = \{a_1, a_2, \dots, a_m\} (m \geq 2)$, and each alternative contains multiple attributes $c_j (j = 1, 2, \dots, n)$. The set of attributes is represented as $C = \{c_1, c_2, \dots, c_n\} (n \geq 2)$, and the set of attributes's weights is denoted by $W = \{\varpi_1, \varpi_2, \dots, \varpi_n\}$, and each attribute's weight satisfies the conditions $\varpi_j \geq 0 (j = 1, 2, \dots, n)$ and $\sum_{j=1}^n \varpi_j = 1$.

3.1. Decision information collection and processing

Suppose that each decision maker $d_k (k = 1, 2, \dots, s)$ represents the decision information of alternative $a_i (i = 1, 2, \dots, m)$ about $c_j (j = 1, 2, \dots, n)$ in linguistic Z- numbers $z_{ij}^k = (A_{ij}^k, B_{ij}^k)$, and the decision information matrix is represented as $Z^k = (z_{ij}^k)_{m \times n} (k = 1, 2, \dots, s)$, that is:

$$Z^k = \begin{pmatrix} z_{11}^k & z_{12}^k & \dots & z_{1n}^k \\ z_{21}^k & z_{22}^k & \dots & z_{2n}^k \\ \vdots & \vdots & \ddots & \vdots \\ z_{m1}^k & z_{m2}^k & \dots & z_{mn}^k \end{pmatrix}, k = 1, 2, \dots, s. \quad (5)$$

On this basis, decision matrix $Z^k = (z_{ij}^k)_{m \times n} (k = 1, 2, \dots, s)$ can be transformed into linguistic scale function matrix $\tilde{Z}^k = (\tilde{z}_{ij}^k)_{m \times n} = [f^*(A_{ij}^k), g^*(B_{ij}^k)]_{m \times n} (k = 1, 2, \dots, s)$ through linguistic scale function (Wang et al., 2024a).

In order to prevent information loss, relative closeness (Wang et al., 2024a) is used in this paper to describe the relative distance between Decision Units (DUS) and ideal and non-ideal solutions, reflecting the closeness of $\tilde{z}_{ij}^k (i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s)$ to ideal and non-ideal solutions, thus converting linguistic Z-numbers into clear numbers and obtaining crisp number decision matrix.

$$\hat{z}_{ij}^k = \frac{d(\tilde{z}_{ij}^k, \hat{z}^-)^2}{d(\tilde{z}_{ij}^k, \hat{z}^+)^2 + d(\tilde{z}_{ij}^k, \hat{z}^-)^2}, i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s. \quad (6)$$

Where $\hat{z}^+$ and $\hat{z}^-$ correspond to the crisp Z-numbers of the best linguistic Z- numbers $\hat{z}^+ = (A_{2g}, B_{2g'})$ and the worst linguistic Z-numbers $\hat{z}^- = (A_0, B_0)$, respectively.

3.2. The decision makers' weights calculation based on group consensus process

In multi-attribute group decision-making, different decision makers often hold varying views on the same issue due to differences in individual cognition, psychological preferences, and decision-making styles. The factors influencing decision makers to adjust their evaluations include not only their assessment of the alternatives but also psychological preferences, confidence in their own opinions, and the

impact of group pressure. Particularly in response to group feedback, decision makers may flexibly adjust their initial assessments depending on the level of confidence or uncertainty they have in their opinions. This process involves complex psychological dynamics, making it difficult for traditional decision-making models to adequately address such situations. If each decision maker rigidly adheres to their initial evaluation, it becomes challenging for the group to reach an effective consensus, leading to reduced decision-making efficiency. Conversely, if decision makers rely too heavily on group opinions, the independence and diversity of individual decisions may be diminished, ultimately affecting the quality of the final decision. To solve this problem, this paper proposes a multi-attribute group consensus model which combines psychological factors and group influences.

3.2.1. A novel psychological preference function

This paper constructs a psychological preference function to describe the psychological tendency and willingness of decision makers to adjust the evaluation. The psychological preference of decision makers affects whether they are willing to modify their initial evaluation according to group feedback, and this adjustment behavior reflects the dynamic adaptability of individuals in the group. The psychological preference function is as follows:

$$\kappa_{ij}^k = \frac{1}{1 + e^{-\chi_k |z_{ij}^k - \hat{z}_{ij}^k|}}, i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s. \quad (7)$$

Where, $z_{ij}^k (i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s)$ is the feedback and modification suggestions given by the top decision maker based on the preliminary evaluation opinions of decision makers, $\hat{z}_{ij}^k (i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s)$ represents the initial evaluation suggestion value of decision maker. $\chi_k (0 \leq \chi_k \leq 1)$ represents the risk avoidance coefficient, which can enable decision makers to achieve an appropriate balance between individual preference and group feedback. $\kappa_{ij}^k (i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s)$ represents the psychological preference level of each decision maker.

Theorem 1. The psychological preference function $\kappa_{ij}^k (i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s)$ is monotonically increasing with respect to the decision makers' adjustment deviation $|z_{ij}^k - \hat{z}_{ij}^k|$.

Proof. Let $\mu_{ij}^k = |z_{ij}^k - \hat{z}_{ij}^k|$, then the psychological preference function is: $\kappa_{ij}^k = \frac{1}{1 + e^{-\mu_{ij}^k \chi_k}}$. By partial differentiation of the function, we have:

$$\frac{d\kappa_{ij}^k}{d\mu_{ij}^k} = \frac{\chi_k \cdot e^{-\mu_{ij}^k \chi_k}}{(1 + e^{-\mu_{ij}^k \chi_k})^2}.$$

Obviously, the first derivative of the function is positive, that is, $\frac{d\kappa_{ij}^k}{d\mu_{ij}^k} \geq 0$. Therefore, the psychological preference function $\kappa_{ij}^k (i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s)$ is monotonically increasing with respect to the decision maker's adjustment deviation $|z_{ij}^k - \hat{z}_{ij}^k|$. $\square$

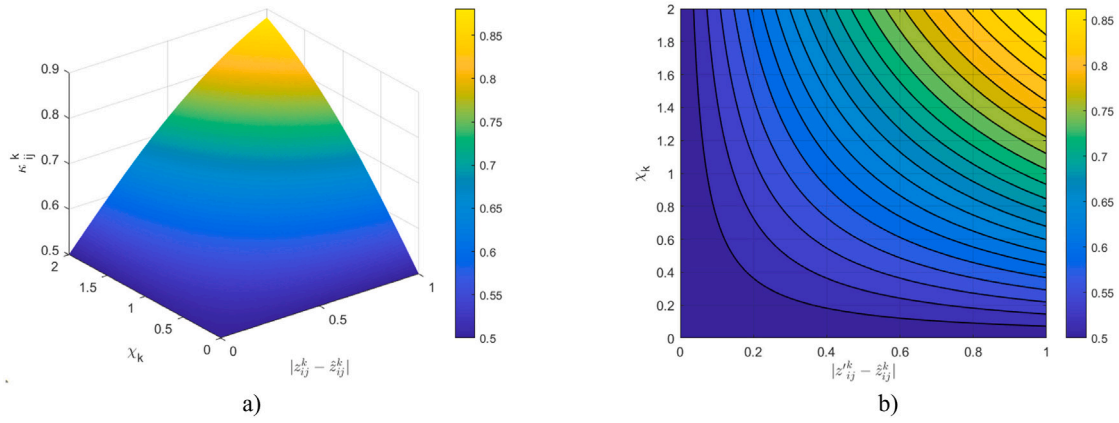


Fig. 2. Simulation analysis of parameter χ_k ($k = 1, 2, \dots, s$) in psychological preference function.

Corollary 1. As the parameter χ_k ($k = 1, 2, \dots, s$) increases, the psychological preference function κ_{ij}^k ($i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s$) becomes more sensitive to the decision makers' adjustment deviation $|z_{ij}^k - \hat{z}_{ij}^k|$.

Proof. Based on the proof of Theorem 1, the second-order partial derivative of the function can be obtained as:

$$\frac{\partial^2 \kappa_{ij}^k}{(\partial \mu_{ij}^k)^2} = \frac{(\chi_k)^2 \cdot e^{-\mu_{ij}^k \cdot \chi_k} \cdot (1 - e^{-\mu_{ij}^k \cdot \chi_k})}{(1 + e^{-\mu_{ij}^k \cdot \chi_k})^3} \geq 0$$

which indicates that the growth rate of the psychological preference function is accelerating. $\square$

Therefore, as the decision maker's adjustment deviation $|z_{ij}^k - \hat{z}_{ij}^k|$ increases, the rate of κ_{ij}^k ($i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s$) not only increases but also accelerates with the increase of χ_k ($k = 1, 2, \dots, s$). The parameter χ_k ($k = 1, 2, \dots, s$) controls the sensitivity of decision makers to bias. A larger value of χ_k ($k = 1, 2, \dots, s$) results in a greater change in the psychological preference function for smaller biases, while a smaller value of χ_k ($k = 1, 2, \dots, s$) makes the bias less influential on the decision makers.

In addition, in order to determine the optimal range of parameters in the psychological preference function, the following simulation analysis is conducted in this paper. Assuming that the value of adjustment deviation $|z_{ij}^k - \hat{z}_{ij}^k|$ belongs to $[0, 1]$, and the value range of χ_k ($k = 1, 2, \dots, s$) belongs to $[0, 2]$. The simulation results are shown in Fig. 2.

According to the simulation analysis results in Fig. 2, when χ_k ($k = 1, 2, \dots, s$) is in the range of 0 to 1, the psychological preference function κ_{ij}^k ($i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s$) shows a relatively stable and effective adjustment characteristic with the increase of χ_k ($k = 1, 2, \dots, s$). At this time, κ_{ij}^k ($i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s$) gradually becomes more sensitive to bias $|z_{ij}^k - \hat{z}_{ij}^k|$ with the increase of χ_k ($k = 1, 2, \dots, s$), which gradually strengthens the response of decision makers to bias, thus effectively prompting decision makers to adjust their preferences in the group decision-making process and further promoting group consensus. However, this change will not lead to overreaction of the system, and a certain degree of balance can be maintained among decision-makers to ensure the effectiveness and consistency of psychological preference adjustment. Therefore, the interval range of $\chi_k \in [0, 1]$ can provide a suitable sensitivity adjustment space, both to accommodate the bias, and to avoid excessive sensitivity response.

On the contrary, when the value of χ_k ($k = 1, 2, \dots, s$) is increased to the interval of 1 to 2, the change of psychological preference function χ_k ($k = 1, 2, \dots, s$) is more drastic, and the response of decision makers

to bias is significantly enhanced. This may lead to overreaction in the system, especially in the adjustment process between decision makers, too high sensitivity will lead to excessive adjustment amplitude, which may cause inconsistency and decision instability between decision makers. This kind of overreaction may lead to the overadjustment of decision makers' psychological preferences and affect the stability and consistency of the final group decision.

Therefore, combined with the sensitivity analysis results of the image, it can be concluded that the interval of $\chi_k \in [0, 1]$ is the most suitable value range. Within this interval, the adjustment effect of psychological preference function can ensure proper feedback and coordination among decision-makers without being overly sensitive, ensure the stability and consistency of group decision-making, and avoid unnecessary interference caused by overreaction.

3.2.2. A group consensus model considering the psychological preferences of decision makers

On the basis, the model accounts for decision makers' confidence in their own evaluations, which affects their flexibility in adjusting: DMs with higher confidence are more likely to maintain their initial judgments, while those with lower confidence are more inclined to accept group feedback. This model simulates the dynamic changes in individual opinions within the group decision-making process, influenced by various psychological and group factors. The detailed framework of the model is outlined in Algorithm 1. And the model constructed in this paper is as follows:

$$\begin{aligned} M_1 : \min & \frac{1}{smn} \sum_{k=1}^s \sum_{i=1}^m \sum_{j=1}^n \kappa_k |z_{ij}^k - z_{ij}^k| + \epsilon \\ \epsilon &= \eta \cdot \sum_{k=1}^s \sum_{i=1}^m \sum_{j=1}^n \max \left\{ 0, \kappa_k \left(|z_{ij}^k - z_{ij}^k| - b_j^k \right) \right\} \quad (8-1) \\ z_{ij}^k &\in [z_{ij}^k - b_j^k, z_{ij}^k + b_j^k] \quad (8-2) \\ \kappa_{ij}^k &= \frac{1}{1 + e^{-\chi_k |z_{ij}^k - \hat{z}_{ij}^k|}} \quad (8-3) \\ s.t. \quad z_{ij}^o &= \sum_{k=1}^s \lambda_k^0 z_{ij}^k \quad (8-4) \\ cl(d_k) &= 1 - \frac{\sum_{i=1}^m \sum_{j=1}^n |z_{ij}^k - z_{ij}^o|}{mn} \quad (8-5) \\ \sum_{k=1}^s \lambda_k cl(d_k) &= CL^* \quad (8-6) \\ 0 &\leq \chi \leq 1, \eta \geq 0, 0 \leq z_{ij}^k \leq 1 \quad (8-7) \\ i &= 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s \quad (8-8) \end{aligned} \quad (8)$$

Where, z_{ij}^o ($i = 1, 2, \dots, m, j = 1, 2, \dots, n$) is the consensus opinion of the group, z_{ij}^k ($i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s$) is the feedback and modification suggestions given by the top decision maker based on the preliminary evaluation opinions of decision makers,

b_j^k ($j = 1, 2, \dots, n, k = 1, 2, \dots, s$) is the confidence degree of each decision maker's preference, and χ_k ($0 \leq \chi_k \leq 1$) represents the risk avoidance coefficient, which can enable decision makers to achieve an appropriate balance between individual preference and group feedback. η ($\eta \geq 0$) is a penalty factor that prevents decision makers from overadjusting, its reasonable setting can balance the relationship between consensus level and adjustment cost in model optimization. In this paper, the empirical setting of $\eta \geq 0$ is used to ensure the constraint validity of the model optimization goal, and in practical applications, the value can be adjusted according to the problem requirements. κ_{ij}^k ($i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s$) represents the psychological preference level of each decision maker, and CL^* represents the maximum group consensus level.

Algorithm 1 Group Consensus Model Considering Decision Makers' Psychological Preferences

Input: Decision maker's initial evaluation opinion $\hat{z}_{ij}^k$ ($i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s$), decision maker's initial weight λ_k^0 ($k = 1, 2, \dots, s$), decision maker's opinion confidence b_j^k ($j = 1, 2, \dots, n, k = 1, 2, \dots, s$), risk avoidance coefficient χ_k ($k = 1, 2, \dots, s$), penalty coefficient η ($\eta \geq 0$).

Output: Feedback z_{ij}^{*k} ($i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s$).

Begin

Step 1: Generate an initial feedback proposal matrix that meets the requirements Z'^k ($k = 1, 2, \dots, s$).

Step 2: For each feedback suggestion matrix candidate solution, calculate the decision-maker's psychological preference level κ_{ij}^k ($i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s$).

Step 3: Calculate adjustment costs.

Step 4: Calculate the group consensus level. If it is higher than the preset threshold, output the feedback suggestion matrix; if it is less than the preset threshold, consider the candidate solution unreasonable and eliminate it.

Step 5: Repeat Steps 2 through 4 and continue optimizing until a solution that meets the consensus requirements is found.

End

The objective function of the model is to minimize the adjustment cost of the group, and there is an additional adjustment cost when the adjustment of the decision maker exceeds the confidence interval. The constraint condition (8–1) reflects the decision-maker's discomfort with excessive deviation from the initial evaluation, and the greater the penalty, the more the model tends to avoid large adjustments. The constraint (8–2) indicates that the elements in each feedback suggestion matrix must be adjusted within a confidence interval of the initial evaluation. The higher the confidence level, the smaller the allowed adjustment, indicating that decision makers are more confident in their initial evaluation and less willing to make adjustments. On the contrary, the lower the confidence level, the easier it is for decision makers to accept the adjustment, and the adjustment range will increase accordingly. The constraints (8–3) are used to describe the degree to which each decision maker is willing to adjust his or her initial evaluation in the face of group feedback. It reflects the psychological tendency of decision makers in the process of adjustment, that is, the balance between insisting on personal preferences and accepting group opinions. The constraints (8–4) - (8–6) are used to calculate the group consensus level. The remaining constraints limit the interval range of each parameter.

Theorem 1. The Model M_1 has an optimal solution.

Proof. Firstly, the objective function of the model is a weighted sum of absolute values and belongs to convex function. According to convex optimization theory, convex functions must have global optimal solutions on compact feasible domains. Secondly, the decision variable z_{ij}^{*k} ($i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s$) is confined to a bounded interval, which ensures the compactness of the feasible solution space, so that the model has an optimal solution within the interval. Finally, all the functions in the constraint condition are continuous and differentiable, which ensures the convergence of the optimization process. Therefore, the optimization problem has an optimal solution under given conditions. $\square$

Since different decision makers may contribute differently to group decision making, taking into account each decision maker's historical contribution can more fairly reflect its impact on decision outcomes. Therefore, in group decision-making, the determination of weights based on the contribution of decision makers can ensure that those decision makers who contribute more to the decision-making process get higher weights (Meng et al., 2023; Liao et al., 2024). This approach not only reflects the quality of current decisions, but also measures the cumulative contribution of decision makers to the long-term decision-making of the group.

$$\phi_k = \sum_{S \subseteq D \setminus \{d_k\}} \frac{|S|! (|D| - |S| - 1)!}{|D|!} [v(S \cup \{d_k\}) - v(S)], k = 1, 2, \dots, s. \quad (9)$$

Where the set S is a subset of the set D that does not contain decision maker d_k ($k = 1, 2, \dots, s$).

At the same time, in order to measure the closeness between a decision maker's current decision and group preferences, an appropriate distance measure is needed. This paper adopts Minkowski distance, which is flexible and can adapt to different decision scenarios by adjusting parameter δ ($\delta > 0$). For example, when $\delta = 1$, Minkowski distance is Manhattan distance, which is suitable for dealing with absolute difference. When $\delta = 2$, the Minkowski distance is the Euclidean distance, which is suitable for dealing with the square difference.

On this basis, this paper combines Shapley value and Minkowski distance to construct a more comprehensive, flexible and reasonable weight calculation function. The Shapley value ensures fairness in each decision maker's historical contribution, while the Minkowski distance precisely measures the decision maker's proximity to group preferences by adjusting for sensitivity to different attributes and types of difference.

$$\lambda_k = \frac{\phi_k \cdot \left(1 / \sum_{i=1}^m \sum_{j=1}^n \omega_j^k |z_{ij}^k - z_{ij}^{\prime 0}|^\delta \right)^{\frac{1}{\delta}}}{\sum_{k=1}^s \phi_k \cdot \left(1 / \sum_{i=1}^m \sum_{j=1}^n \omega_j^k |z_{ij}^k - z_{ij}^{\prime 0}|^\delta \right)^{\frac{1}{\delta}}}, k = 1, 2, \dots, s. \quad (10)$$

Where δ ($\delta > 0$) represents the order of Minkowski and ω_j^k ($k = 1, 2, \dots, s, j = 1, 2, \dots, n$) represents the initial attribute weight of each decision maker.

Theorem 2. The weight λ_k ($k = 1, 2, \dots, s$) of the decision maker d_k ($k = 1, 2, \dots, s$) satisfies the sum of accumulation is 1, that is: $\sum_{k=1}^s \lambda_k = 1$.

Proof. The sum of k from 1 to s yields:

$$\sum_{k=1}^s \lambda_k = \sum_{k=1}^s \frac{\phi_k \cdot \left(\frac{1}{\sum_{i=1}^m \sum_{j=1}^n \omega_j^k |z_{ij}^k - z_{ij}^{\prime 0}|^\delta} \right)^{\frac{1}{\delta}}}{\sum_{k=1}^s \phi_k \cdot \left(\frac{1}{\sum_{i=1}^m \sum_{j=1}^n \omega_j^k |z_{ij}^k - z_{ij}^{\prime 0}|^\delta} \right)^{\frac{1}{\delta}}},$$

so it can be simplified to $\sum_{k=1}^s \lambda_k = \frac{\sum_{k=1}^s \alpha_k}{\sum_{k=1}^s \alpha_k} = 1$, where $\alpha_k = \phi_k \cdot \left(1 / \sum_{i=1}^m \sum_{j=1}^n \omega_j^0 |z'_{ij}{}^k - z''_{ij}{}^o|^\delta\right)^{\frac{1}{\delta}}$. $\square$

Theorem 3. The smaller the distance between decision matrix Z'^k and group preference matrix Z''^o , the greater the weight λ_k ($k = 1, 2, \dots, s$) of decision maker d_k ($k = 1, 2, \dots, s$).

Proof. Firstly, define D_k ($k = 1, 2, \dots, s$) as the distance between decision maker d_k ($k = 1, 2, \dots, s$) and group preferences:

$D_k = \sum_{i=1}^m \sum_{j=1}^n \omega_j^0 |z'_{ij}{}^k - z''_{ij}{}^o|^\delta$, so $\lambda_k = \frac{\phi_k \cdot D_k^{-\frac{1}{\delta}}}{\sum_{k=1}^s \phi_k \cdot D_k^{-\frac{1}{\delta}}}$. Secondly, the first-order derivative is calculated for this function: $\frac{\partial \lambda_k}{\partial D_k} = \frac{-\frac{1}{\delta} \phi_k \cdot (D_k)^{-\frac{1}{\delta}-1} \cdot \sum_{k=1}^s \phi_k (D_k)^{-\frac{1}{\delta}}}{\left(\sum_{k=1}^s \phi_k (D_k)^{-\frac{1}{\delta}}\right)^2} < 0$, so the smaller the distance

D_k ($k = 1, 2, \dots, s$) between the decision maker and the group preference, the greater the decision makers' weights λ_k ($k = 1, 2, \dots, s$). $\square$

3.3. The weights calculation of the attributes considering the synergy effect

In the multi-attribute group decision making problem, it is difficult to accurately define the relative importance of each attribute and fully reflect the marginal contribution of each attribute to the overall decision utility because of the difference in the weight allocation of attributes by different decision makers. In order to solve the above problems, this paper constructs an optimization model of attribute weights based on Shapley values in cooperative game theory. The model not only relies on Shapley values to calculate attributes' weights (Liao et al., 2024; Wu et al., 2023), but also comprehensively considers the marginal contributions of each attribute in different subset combinations, and takes into account fairness and utility maximization in the weight allocation process. By optimizing the weight allocation scheme, it ensures that the contribution of each attribute to the overall decision utility not only reflects its actual importance, but also conforms to the principle of fair distribution, so as to realize the optimal equilibrium of the overall decision utility. The optimization model is as follows:

$$M_2 : \max \sum_{j=1}^n \omega_j \phi_j$$

$$\begin{cases} \sum_{j=1}^n \omega_j = 1, \omega_j \geq 0 & (11-1) \\ \phi_j = \sum_{S \subseteq C \setminus \{c_j\}} \frac{|S|!(n-|S|-1)!}{n!} (v(S \cup \{c_j\}) - v(S)) & (11-2) \\ s.t. \begin{cases} \phi_j \geq 0 & (11-3) \\ v(S \cup \{c_j\}) - v(S) \geq 0 & (11-4) \\ v(S) = \sum_{\{c_j\} \in S} \ln \left(\sum_{i=1}^m z''_{ij}{}^o \right) & (11-5) \\ j = 1, 2, \dots, n & (11-6) \end{cases} \end{cases} \quad (11)$$

Where the set S is a subset of the set C that does not contain attribute c_j ($j = 1, 2, \dots, n$). ω_j ($j = 1, 2, \dots, n$) is the weight corresponding to each attribute, and $z''_{ij}{}^o$ ($i = 1, 2, \dots, m, j = 1, 2, \dots, n$) represents the group consensus opinion aggregated by decision-makers based on the feedback suggestion value.

The objective function of the model is to weight and maximize the weight of each attribute and the Shapley value, that is, by adjusting the weight of each attribute, the contribution of each attribute is matched with its importance, and the situation that some attributes are overestimated or underestimated is avoided, so as to maximize the overall group utility. The Shapley value in the objective function reflects the marginal contribution of each attribute in the MAGDM,

and the goal is to find an optimal weight distribution, so that the contribution of all attributes can be reasonably balanced, so as to maximize the group utility. The constraints of the model mainly include the following aspects: Constraint condition (11-1) requires the sum of all attributes' weights to be 1, and each attributes' weights must be non-negative. The constraint (11-2) defines the calculation of the Shapley value, a formula that calculates the marginal contribution of each attribute by traversing all possible subsets of attributes. The constraint (11-3) ensures that the Shapley value must be non-negative, reflecting that the marginal utility of each attribute should not be negative, i.e. the addition of each attribute should contribute at least to the overall utility. The constraint condition (11-4) requires that the marginal contribution of attributes must be greater than or equal to zero, which ensures that the increase of each attribute is always beneficial, avoiding the situation that the addition of some attributes reduces the effectiveness of group decision making. The constraints (11-5) define a concrete form of the utility function that calculates the utility of a subset by logarithmically transforming the evaluations of the alternative under each attribute and summing the evaluations of each attribute.

Theorem 4. The Model M_2 has an optimal solution.

Proof. Firstly, the objective function of this model is a linear combination of attributes' weights and the weighted sum of Shapley values, and all constraints are linear. Secondly, all the constraints form a closed and bounded set of feasible solutions. In particular, the constraints include a sum of attributes' weights of 1 and non-negative constraints, ensuring that all solutions lie within a bounded region. In addition, Shapley value is used as the measure of attribute contribution degree, and its calculation formula ensures the non-negative and continuity of contribution degree. Therefore, the objective function is continuous in this bounded feasible region. According to Weierstrass extreme value theorem, there must be an optimal solution for a continuous function in a bounded closed region. Since the objective function of this model is linear, and all the constraints are linear, it can be guaranteed that the model M_2 has an optimal solution. under the given constraints. $\square$

3.4. An approach to ranking alternatives based on quantum probability

In the complex emergency decision-making process, the mutual influence and interference effect among decision makers cannot be ignored. This paper proposes a three-layer quantum-like Bayesian network structure to assist emergency teams in making informed choices within the complex, multi-decision maker, and multi-attribute environment by modeling the relationships between decision makers, attribute factors, and emergency alternatives.

The structure of quantum-like Bayesian network consists of three layers: the first layer is the decision-maker layer, which represents the decision preference and mutual influence of different participants; The second layer is the attribute layer, which covers the key factors affecting the decision, such as disaster risk, resource allocation, rescue ability, etc. The third layer is the scenario layer, which shows the different emergency scenarios that need to be evaluated. In this network, not only the independent evaluation of each decision maker to the scheme is considered, but also the interference effect among decision makers is included, and the influence of the interaction of opinions between decision makers on the final decision is reflected through the quantum mechanism. By calculating the quantum probability value of each alternative, the network is able to rank the alternatives and dynamically update the influence among decision makers, thus providing more scientific support for complex decision problems.

(1) The weights of the decision makers constitutes the first layer probability of the QLBN.

$$P(d_k) = \lambda_k, k = 1, 2, \dots, s. \quad (12)$$

(2) The conditional probability is calculated as follows:

$$P(a_i|d_k) = \frac{\sum_{j=1}^n \omega_j z'_{ij}}{\sum_{i=1}^m \sum_{j=1}^n \omega_j z'_{ij}}, i = 1, 2, \dots, m, k = 1, 2, \dots, s. \quad (13)$$

(3) To calculate the cosine of the interference phase angle of the decision maker, the specific steps are as follows:

Firstly, the preference matrix $Q = (q_{ik})_{m \times s}$ of decision makers for alternatives is obtained based on Bayesian networks. q_{ik} represents decision makers $d_k (k = 1, 2, \dots, s)$ preference for alternative $a_i (i = 1, 2, \dots, m)$, and if $\sum_{i=1}^m \sum_{k=1}^s q_{ik} = 1$ is satisfied, the calculation is as follows:

$$q_{ik} = P(d_k)P(a_i|d_k), i = 1, 2, \dots, m, k = 1, 2, \dots, s. \quad (14)$$

On this basis, according to the description in literature (She et al., 2021; Han and Liu, 2023), as an entropy measure in quantum information theory, Deng entropy has advantages in dealing with uncertainty and fuzzy information, and is especially suitable for decision-maker preference modeling in group decision-making scenarios. Because it can effectively characterize the mutual interference or synergy between decision makers' preference vectors. Compared with von Neumann entropy, Deng entropy has better adaptability in dealing with discrete probability distributions, especially the subjective evaluation of decision makers in group decision making, and can more accurately describe the uncertainty of information and the interaction between decision makers. Secondly, calculations of von Neumann entropy are prone to zero interference, especially when there is no obvious correlation or interference between system states or preference vectors. In the framework of quantum probability, von Neumann entropy is based on density matrix, which is suitable for describing the mixing degree of quantum states. However, when the difference between preference vectors is very small or the decision makers converge, von Neumann entropy can result in calculations that are close to zero and cannot effectively account for interference effects or subtle differences between decision makers. By introducing the measure of probability distribution, Deng entropy can still maintain high sensitivity when dealing with preference vectors with high similarity or certain overlap, thus effectively avoiding the "zero interference" problem that may appear in von Neumann entropy. Therefore, Deng entropy is used to measure the interference between decision makers in this paper.

Finally, the probability of each alternative is calculated:

$$\begin{aligned} p(a_i) &= \sigma \left| \sum_{k=1}^s \varphi_{d_k} e^{i\theta_{d_k}} \varphi_{a_i|d_k} e^{i\theta_{a_i|d_k}} \right|^2 \\ &= \sigma \left[\sum_{k=1}^s |\varphi_{d_k} \varphi_{a_i|d_k}|^2 + 2 \sum_{k=1}^{s-1} \sum_{\hat{k}=k+1}^s \varphi_{d_k} \varphi_{a_i|d_k} \varphi_{d_{\hat{k}}} \varphi_{a_i|d_{\hat{k}}} \cos(\theta_{ik} - \theta_{i\hat{k}}) \right] \\ &= \sigma \left[\sum_{k=1}^s |\varphi_{d_k} \varphi_{a_i|d_k}|^2 + interference_i \right], i = 1, 2, \dots, m. \end{aligned} \quad (15)$$

Where,

$$\sigma = \frac{1}{\sum_{i=1}^m \left[\sum_{k=1}^s |q_{ik}|^2 + 2 \sum_{k=1}^{s-1} \sum_{\hat{k}=k+1}^s q_{ik} q_{i\hat{k}} \cos(\theta_{ik} - \theta_{i\hat{k}}) \right]^2}. \quad (16)$$

3.5. Decision steps

In this paper, a multi-attribute group consensus model is proposed, which adjusts the consensus under the quantum framework, and considers the interference effect between decision makers and their psychological preferences. The model framework is shown in Fig. 3. The construction of the model includes the following steps:

Step1: Determine the evaluation information. Each decision maker $d_k (k = 1, 2, \dots, s)$ evaluates attribute $c_j (j = 1, 2, \dots, n)$ in alternative $a_i (i = 1, 2, \dots, m)$ and obtains the initial information evaluation matrix $Z^k = (z_{ij}^k)_{m \times n} (k = 1, 2, \dots, s)$. The evaluation information matrix is transformed into the linguistic scale function matrix $\tilde{Z}^k = (\tilde{z}_{ij}^k)_{m \times n} (k = 1, 2, \dots, s)$ according to the linguistic scale function, and the decision matrix $\hat{Z}^k = (\hat{z}_{ij}^k)_{m \times n} (k = 1, 2, \dots, s)$ according to Eq. (6).

Step 2: Decision makers' weights determination based on group consensus process. To solve the model (8), the feedback suggestion matrix of each decision maker is obtained, and the weight of each decision maker is calculated by Eqs. (9)–(10).

Step 3: Combine the synergistic effect between attributes to determine the attributes' weights. The optimization model (11) based on Shapley is solved to obtain the optimal attributes' weights.

Step 4: Determine the quantum-like Bayesian network (QLBN). The weight $\lambda_k (k = 1, 2, \dots, s)$ of the decision maker is determined as the node probability $P(d_k) (k = 1, 2, \dots, s)$ in the first layer of the QLBN. The conditional probability $P(a_i|d_k) (i = 1, 2, \dots, m, k = 1, 2, \dots, s)$ in QLBN is obtained from Eq. (13).

Step 5: Calculate the quantum probabilities of the alternatives. The Eqs. (14)–(16) is used to calculate the quantum probabilities of alternatives, and the alternatives are sorted according to the quantum probability value.

4. Case analysis

In this section, a concrete example is used to verify the effectiveness and superiority of the model in dealing with the interaction between decision makers and the correlation between attributes.

4.1. Background

In September 2024, Typhoon Bebinca formed and gradually strengthened in the South China Sea, and is expected to make landfall in the Nanjing area in late September. Bebinca was one of the strongest typhoons to threaten Nanjing in recent years, with heavy rainfall, high wind speeds and the potential risk of causing flash floods and water-logging. According to the forecast of the meteorological department, the typhoon may be accompanied by short-term heavy rainfall when it makes landfall, resulting in serious water accumulation in low-lying areas of the city, and high wind speed may cause damage to the city's infrastructure, especially the power supply and transportation system. The approaching typhoon necessitated the Nanjing Municipal government to make emergency plans quickly to reduce the damage to the city and ensure the safety of citizens' lives and property. The Nanjing emergency management department has set up an emergency decision-making team, which includes experts from the meteorological department, officials from the emergency management bureau and community representatives. The team will need to evaluate five alternatives $a_i (i = 1, 2, \dots, 5)$ in a short period of time to determine which alternative will be most effective in responding to a typhoon and minimizing disaster risk.

Decision team members: (1) Meteorological expert d_1 : mainly provide professional prediction of typhoon track, intensity and impact, and give opinions on the accuracy of typhoon warning and the control of time nodes. (2) Emergency management Officer d_2 : responsible for assessing operational issues such as resource allocation, evacuation arrangement and emergency material preparation in each plan to ensure the execution and feasibility of the emergency plan. (3) Community representative d_3 : responsible for assessing the actual effects of the emergency plan from the perspective of the residents, especially the rationality of evacuation and refuge measures, the choice of resettlement sites and the acceptance of the plan by the community.

The attributes of the alternative are as follows: (1) Accuracy of typhoon warning c_1 . The timing and accuracy of typhoon warnings are critical to the overall emergency response plan. Accurate track and

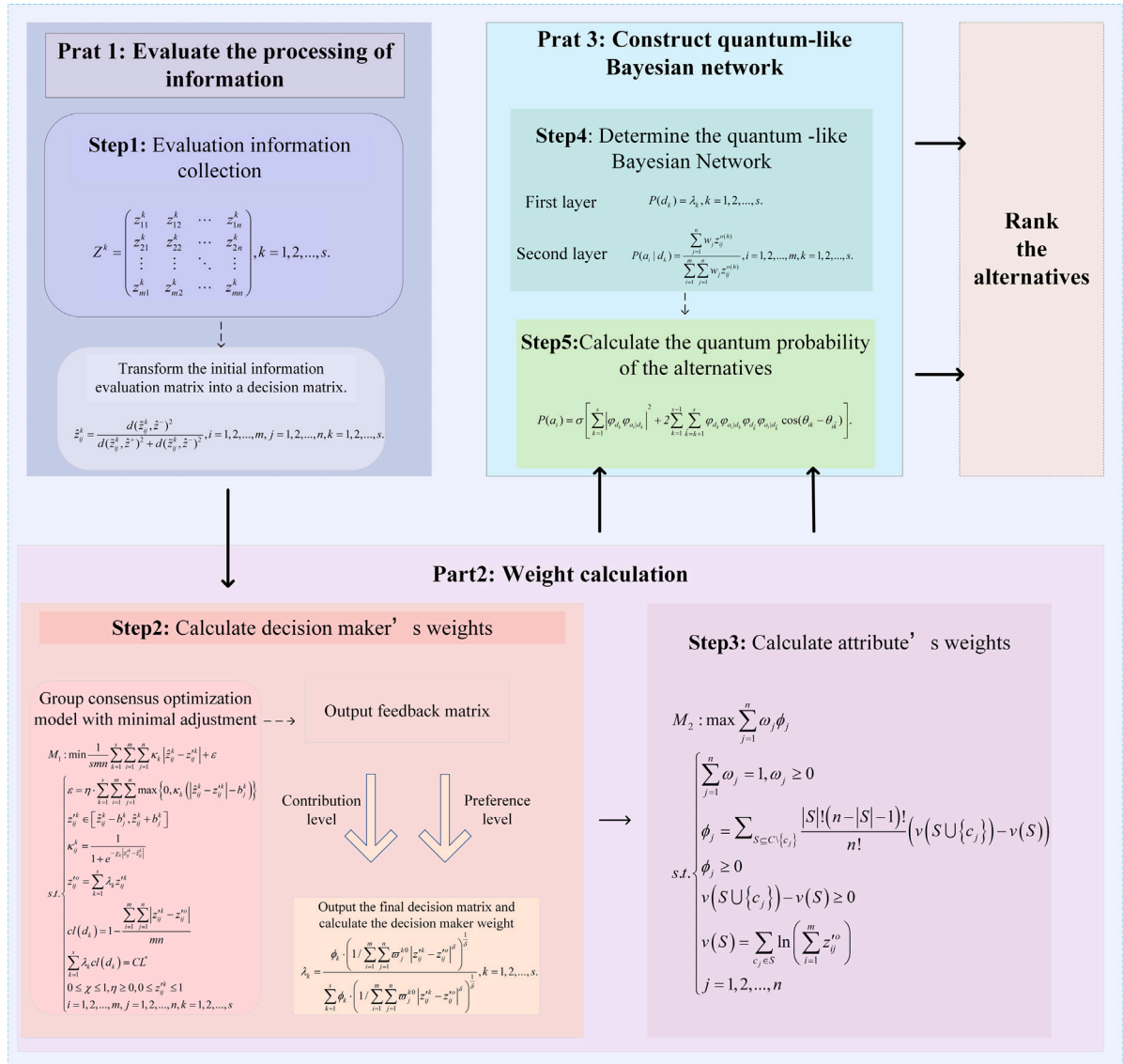


Fig. 3. The framework of the method proposed in this paper.

rainfall forecasts can help governments deploy resources and evacuate residents in advance. The attribute includes the advance of warning time, prediction accuracy, and the propagation speed of warning information. (2) Evacuation and refuge measures c_2 . This attribute assesses the feasibility of evacuation plans and the placement of residents. This includes the planning of evacuation routes, traffic management measures, the number and conditions of shelters (e.g. windproof, water-proofing, etc.), and special arrangements for vulnerable groups (e.g. the elderly, children). The merits of the scheme are reflected in the rapid and orderly evacuation of residents to safe shelters. (3) Material reserve and distribution c_3 . Prior to a disaster, the stockpiling and distribution of critical emergency supplies (such as drinking water, food, medical supplies, etc.) is key to determining disaster response capacity. This attribute evaluates the reserve amount of materials, the mode of transportation, the rationality of the distribution mechanism, and the timely delivery of emergency materials to the severely affected areas after the typhoon. (4) Infrastructure protection c_4 . Nanjing's power system, communication network, transportation network and other infrastructure is the most vulnerable part of the typhoon damage. This attribute assesses how each option protects critical infrastructure, such

as whether to reinforce power facilities, protect communication towers, and strengthen drainage systems to reduce the impact of waterlogging caused by storms on traffic. (5) Post-disaster recovery ability c_5 . After a typhoon, restoring urban functions quickly is the key to mitigating the impact of a disaster. This attribute includes speed of restoration of water and electricity supplies, road and bridge repair, garbage and waste removal, reconstruction planning, etc. Assessing the response capacity of each program during the post-disaster recovery period determines whether the city can return to normal order as soon as possible.

4.2. Decision steps

Step 1: Collect decision information. The two linguistic sets of this case are: $A = \{A_0 : \text{Very poor}, A_1 : \text{poor}, A_2 : \text{slightly poor}, A_3 : \text{fair}, A_4 : \text{slightly good}, A_5 : \text{good}, A_6 : \text{very good}\}$, $B = \{B_0 : \text{Extremely Unlikely}, B_1 : \text{Unlikely}, B_2 : \text{Moderate}, B_3 : \text{Likely}, B_4 : \text{Extremely Likely}\}$.

Therefore, the decision matrix of three decision makers using linguistic Z-numbers to represent decision information is shown below.

$$Z^1 = \begin{bmatrix} (A_3, B_2) & (A_1, B_4) & (A_0, B_0) & (A_0, B_1) & (A_2, B_1) \\ (A_4, B_2) & (A_3, B_2) & (A_3, B_4) & (A_2, B_0) & (A_4, B_3) \\ (A_6, B_3) & (A_5, B_3) & (A_4, B_3) & (A_6, B_4) & (A_2, B_4) \\ (A_3, B_4) & (A_4, B_3) & (A_2, B_4) & (A_2, B_3) & (A_3, B_2) \\ (A_2, B_2) & (A_1, B_3) & (A_4, B_0) & (A_5, B_2) & (A_4, B_3) \\ (A_4, B_2) & (A_5, B_3) & (A_2, B_1) & (A_2, B_2) & (A_2, B_1) \\ (A_0, B_0) & (A_6, B_3) & (A_1, B_3) & (A_3, B_3) & (A_3, B_4) \\ (A_6, B_4) & (A_4, B_4) & (A_1, B_2) & (A_5, B_3) & (A_4, B_2) \\ (A_2, B_1) & (A_3, B_1) & (A_5, B_3) & (A_6, B_4) & (A_3, B_2) \\ (A_3, B_2) & (A_1, B_4) & (A_2, B_2) & (A_3, B_4) & (A_2, B_3) \\ (A_2, B_3) & (A_2, B_2) & (A_4, B_2) & (A_3, B_1) & (A_1, B_1) \\ (A_1, B_4) & (A_6, B_3) & (A_3, B_2) & (A_4, B_2) & (A_0, B_0) \\ (A_3, B_2) & (A_3, B_2) & (A_2, B_1) & (A_3, B_3) & (A_2, B_4) \\ (A_4, B_1) & (A_3, B_3) & (A_6, B_2) & (A_6, B_4) & (A_5, B_2) \\ (A_3, B_3) & (A_2, B_3) & (A_4, B_3) & (A_5, B_2) & (A_1, B_1) \end{bmatrix},$$

Step 2: This paper assumes that the confidence of each decision-maker for each attribute is 0.1. On this basis, the feedback suggestion matrix Z'^k ($k = 1, 2, \dots, 3$) provided by the top decision-maker to each decision-maker can be obtained by solving model (8).

$$Z'^1 = \begin{bmatrix} 0.8692 & 0.6183 & 0.0321 & 0.2419 & 0.3398 \\ 0.9165 & 0.7019 & 0.7482 & 0.1308 & 0.7776 \\ 0.8626 & 0.8803 & 0.7730 & 0.9646 & 0.6990 \\ 0.8258 & 0.9478 & 0.7500 & 0.5330 & 0.8590 \\ 0.5194 & 0.5409 & 0.2909 & 0.9236 & 0.7607 \\ 0.9510 & 0.8809 & 0.5554 & 0.3147 & 0.3282 \\ 0.2117 & 0.9685 & 0.5757 & 0.7579 & 0.6976 \\ 0.9661 & 0.9830 & 0.8427 & 0.9633 & 0.8772 \\ 0.3823 & 0.3918 & 0.9573 & 0.9300 & 0.6341 \\ 0.7183 & 0.5909 & 0.4215 & 0.8205 & 0.7468 \\ 0.7816 & 0.5641 & 0.8665 & 0.6416 & 0.2339 \\ 0.7500 & 0.9273 & 0.5659 & 0.8540 & 0.0442 \\ 0.4824 & 0.6988 & 0.4305 & 0.8715 & 0.6839 \\ 0.7745 & 0.7032 & 0.9811 & 0.8713 & 0.9410 \\ 0.6629 & 0.7563 & 0.8793 & 0.8940 & 0.4139 \end{bmatrix},$$

On this basis, the group consensus level is calculated as 0.8538, which is higher than the group consensus threshold of 0.85. Therefore, the decision-makers' weights is calculated according to Eqs. (9)–(10).

$$A = \{0.3510, 0.3565, 0.2925\}.$$

Step 3: Calculate the attributes's weights. Based on the weight of each decision maker and the adjustment matrix of each decision maker, OWA operator is used to obtain the group aggregation matrix.

$$Z'^0 = \begin{bmatrix} 0.8727 & 0.6961 & 0.4627 & 0.3848 & 0.3047 \\ 0.6165 & 0.8629 & 0.6334 & 0.5659 & 0.5346 \\ 0.7883 & 0.8638 & 0.6977 & 0.9369 & 0.7563 \\ 0.6527 & 0.6784 & 0.8915 & 0.7735 & 0.8028 \\ 0.6323 & 0.6217 & 0.5096 & 0.8782 & 0.6543 \end{bmatrix}.$$

On this basis, model (11) is solved to obtain the optimal weights of attributes.

$$W = \{0.2179, 0.1371, 0.2049, 0.2088, 0.2313\}.$$

Step 4: Set the decision-maker weight to the first layer of the QLBN and calculate the conditional probability according to Eq. (12).

$$\begin{aligned} P(a_1|d_1) &= 0.1247, & P(a_2|d_1) &= 0.1998, & P(a_3|d_1) &= 0.2526, \\ P(a_4|d_1) &= 0.2355, & P(a_5|d_1) &= 0.1874, & P(a_1|d_2) &= 0.1685, \\ P(a_2|d_2) &= 0.1780, & P(a_3|d_2) &= 0.2659, & P(a_4|d_2) &= 0.1947, \\ P(a_5|d_2) &= 0.1929, & P(a_1|d_3) &= 0.1796, & P(a_2|d_3) &= 0.1743, \\ P(a_3|d_3) &= 0.1843, & P(a_4|d_3) &= 0.2536, & P(a_5|d_3) &= 0.2082. \end{aligned}$$

Finally, the quantum probabilities of alternatives as follows:

$$\begin{aligned} P(a_1) &= 0.1384, & P(a_2) &= 0.1909, & P(a_3) &= 0.1416, \\ P(a_4) &= 0.3079, & P(a_5) &= 0.2212. \end{aligned}$$

And the alternatives are ranked as: $a_4 > a_5 > a_2 > a_3 > a_1$.

4.3. Sensitivity analysis

(1) The influence of psychological preference factor χ_k ($k = 1, 2, 3$) on consensus level

In the process of group consensus, the psychological preferences of different decision makers have an important influence on the formation of group consensus. Since each decision maker has different willingness to adjust in the face of group feedback, the psychological preference parameter χ_k ($k = 1, 2, 3$) will directly affect whether decision makers are willing to deviate from their initial judgment and approach the group consensus, as shown in Fig. 4.

As can be seen from Fig. 4, the consensus level of decision maker d_1 fluctuates significantly with the change of psychological preference parameter χ_k ($k = 1, 2, 3$), indicating that decision maker d_1 is highly sensitive to psychological factors. When χ_k ($k = 1, 2, 3$) is low, the consensus level of decision maker d_1 is high, but with the increase of χ_k ($k = 1, 2, 3$), the consensus level fluctuates significantly. The consensus level of decision maker d_2 and decision maker d_3 is relatively stable, and the fluctuation range is small with the change of χ_k ($k = 1, 2, 3$), indicating that they are less sensitive to psychological preference factors. On the whole, the consensus level of the group remained relatively stable under different values of χ_k ($k = 1, 2, 3$) and did not show significant fluctuations, indicating that although individual psychological factors can affect individual adjustment willingness, they have limited impact on the overall consensus of the group.

Through this sensitivity analysis, it can be seen that there are significant differences in the sensitivity of different decision makers to psychological preference factors, which indicates that the adjustment willingness of individuals plays an important role in the formation of consensus in the multi-attribute group decision-making process. For those decision-makers who are more sensitive to psychological preferences, group consensus is more challenging to reach, while decision-makers who are not sensitive to psychological factors are more likely to reach agreement with the group.

(2) The influence of confidence b_j^k ($k = 1, 2, 3, j = 1, 2, \dots, 5$) on consensus level

In the process of multi-attribute group decision making, the decision maker's confidence in his own evaluation greatly affects his willingness to adjust the group feedback. Decision-makers with high levels of confidence tend to stick to their initial opinions more, while decision-makers with low levels of confidence are more receptive to group opinions. As shown in Fig. 5.

As can be seen from Fig. 5, with the change of confidence, the consensus level of the group as a whole remains relatively stable, with a small fluctuation range between 0.83 and 0.85. Individual decision makers react differently to changes in confidence. The consensus level of decision maker d_1 is highly sensitive to the change of confidence, reaching d_1 peak when b_j^1 ($j = 1, 2, \dots, 5$) = 0.2, and then decreasing significantly with the increase of confidence, showing the persistence of high confidence. However, the consensus level of decision makers d_2 and d_3 showed a more stable trend in the whole confidence change process, and the fluctuation range was small, and it remained at a high level, indicating that they could better maintain consistency with the group under different confidence levels.

The change of confidence significantly affects the consensus level of individual decision-makers, especially for those decision-makers who are more sensitive to confidence, the increase of confidence will cause them to stick to their initial opinions, and thus weaken the formation of group consensus. For decision-makers with low or insensitive

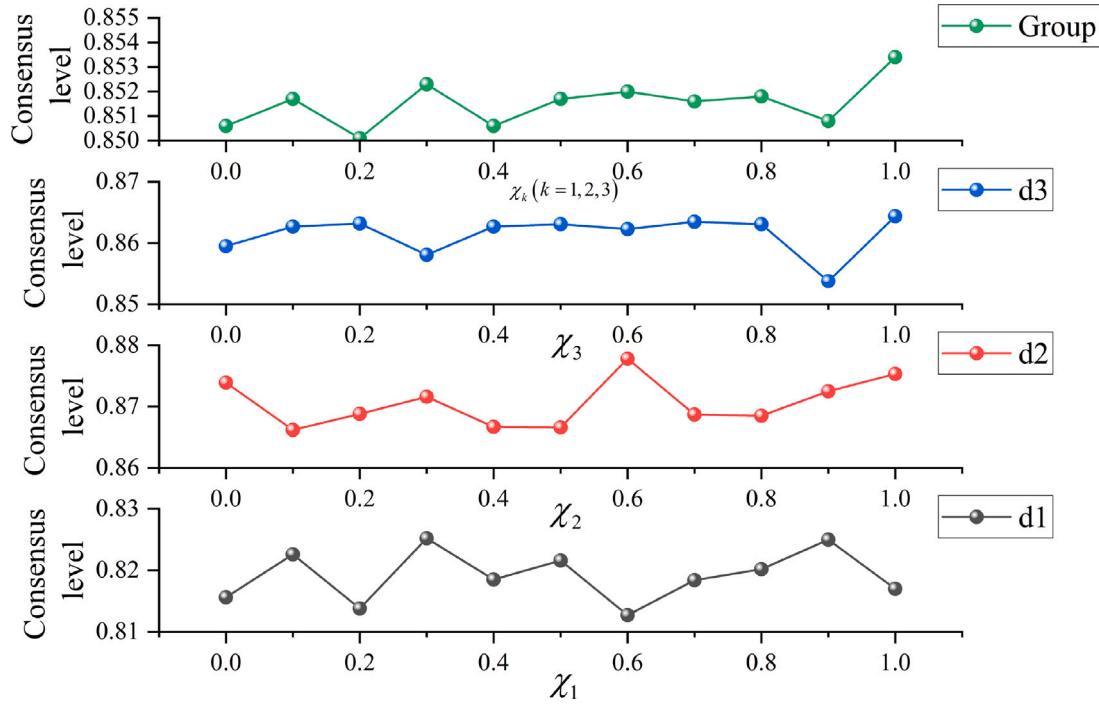


Fig. 4. Influence of psychological preference factors χ_k ($k = 1, 2, 3$) on consensus level.

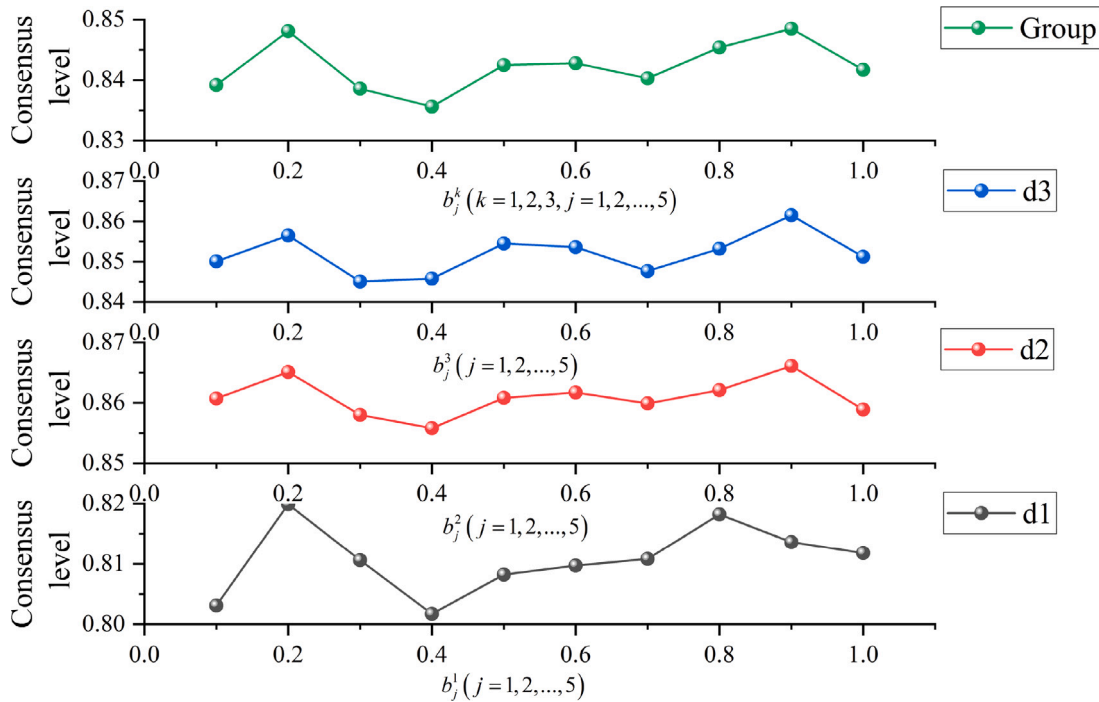


Fig. 5. Influence of confidence b_j^k ($k = 1, 2, 3, j = 1, 2, \dots, 5$) on group consensus level.

confidence, they are more receptive to the opinions of the group and show a higher level of consensus. Therefore, in multi-attribute decision-making, rationally guiding decision-makers with different confidence levels to accept group opinions under appropriate circumstances can effectively improve the overall consensus level of the group and optimize the decision result.

4.4. Simulation analysis

In multi-attribute group decision-making, the number of decision makers is a critical factor influencing the formation of group consensus. As the number of decision makers changes, the mechanisms of information exchange and feedback are also affected, which directly impacts

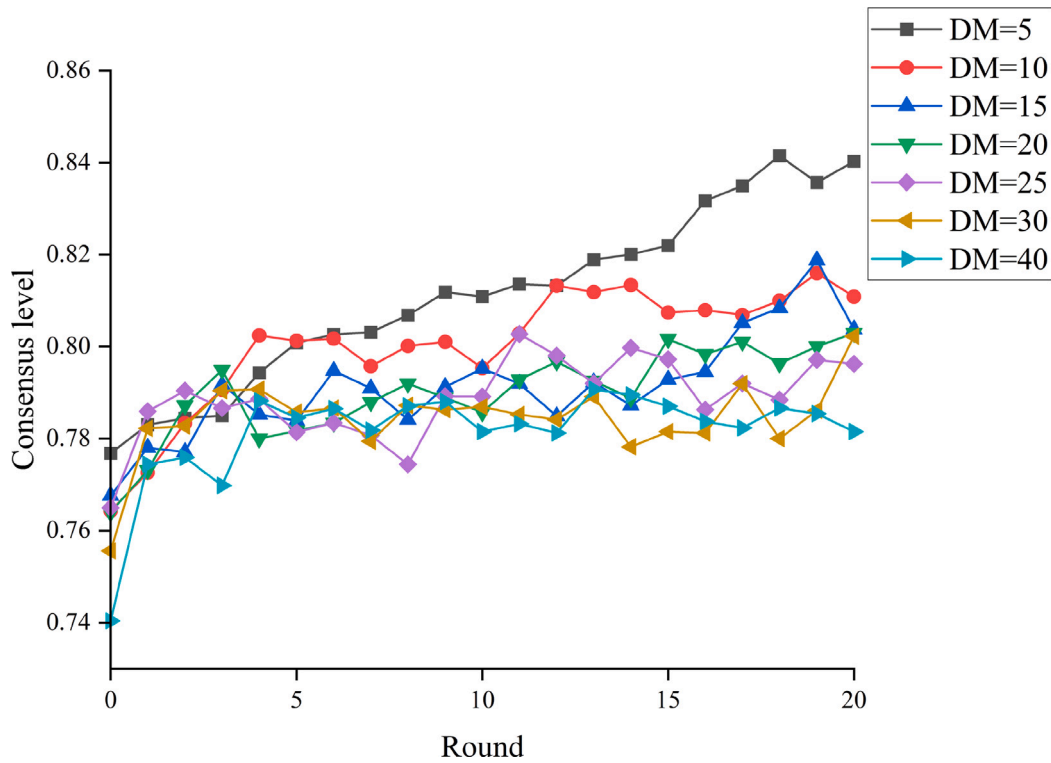


Fig. 6. Efficiency effect of the number of decision makers on the group consensus level.

the group's level of consensus. An increase in decision makers may lead to greater divergence of opinions, making it more difficult to achieve consensus. Conversely, a smaller number of decision makers can reduce information diversity, potentially lowering the quality of the consensus.

This paper conducted sensitivity analysis by randomly generating different numbers of decision matrices (e.g., 5, 10, 15, 20, 25, 30, 40 decision makers) and applying them to the group consensus model. The analysis aimed to evaluate the model's performance in terms of consensus formation across different group sizes, and to verify its effectiveness and robustness with varying numbers of decision makers, as shown in Fig. 6.

As shown in Fig. 6, the number of decision makers significantly influences the group consensus level over multiple decision-making rounds. When the number of decision makers is small (e.g., DM = 5 and DM = 10), the group achieves a high consensus level quickly and remains relatively stable. In particular, the group with DM = 5 exhibited a rapid increase in consensus in the initial rounds, followed by a slower rise, eventually stabilizing at around 0.85. This suggests that fewer decision makers tend to reach a higher consensus more quickly.

By contrast, as the number of decision makers increases (e.g., DM = 15 to 40), the consensus level becomes more volatile. In these cases, the group's consensus typically fluctuated between 0.77 and 0.80, with significant variations across rounds. This indicates that a larger number of decision makers increases the complexity of information exchange, making it harder for the group to reach and maintain consensus. As the number of decision makers grows, divergent opinions become more prominent, leading to greater fluctuations in consensus during subsequent rounds.

Despite these variations, the model demonstrates a consistent trend in response to different group sizes. Although the number of decision makers has a notable effect on the consensus level, the model maintains a reasonable consensus across all scenarios. This further validates the model's effectiveness and robustness, with simulation results confirming its applicability and stability in diverse decision-making contexts.

4.5. Comparative analysis

In order to verify the effectiveness of the proposed method, several existing decision models are compared and analyzed, and the final ranking results obtained by different methods are shown in Table 2.

It can be concluded from Table 2 that the proposed method has a significant difference in the ranking of alternatives compared to the traditional decision model. For example, the TOPSIS (Zhang and Li, 2023a), TODIM (Llamazares, 2018) and VIKOR (Mishra et al., 2022) methods all regard a_3 as the optimal alternative, but the optimal alternative obtained by the method proposed in this paper is a_4 . Specifically, TOPSIS (Zhang and Li, 2023a) method determines the pros and cons of alternative solutions by calculating the distance between each alternative solution and the ideal solution and the negative ideal solution; TODIM (Llamazares, 2018) method uses a weighted distance measure to sort alternative solutions; VIKOR (Mishra et al., 2022) method comprehensively sorts alternative solutions by combining different evaluation information. The common feature of these methods is that they assume that the preferences and attributes of decision makers are independent, and the mutual interference effect among group decision makers is usually not fully considered in the decision-making process, resulting in the bias of individual decision makers in group decision making, thus affecting the final ranking result. In addition, although MAGDM (Zha et al., 2021) introduced the group consensus adjustment mechanism, it did not consider the psychological state of decision makers and the mutual interference effect among decision makers. This model mainly focuses on how to achieve group consensus by adjusting the evaluation information of decision makers, but it fails to effectively capture the psychological reaction of decision makers in the face of group pressure, nor does it introduce quantum probability to describe the interference effect among decision makers. Therefore, MAGDM model (Zha et al., 2021) may have certain limitations when dealing with complex decision-making scenarios with high uncertainty and interference from decision makers, which may easily lead to decision results with large deviations. At the same time, when the model proposed in this paper degenerates into a classical

Table 2
Comparison of ranking results of different models.

Methods	Ranking indices	Ranking results
TOPSIS (Zhang and Li, 2023a)	$C_1^+ = 0.2147, C_2^+ = 0.3976, C_3^+ = 0.7651,$ $C_4^+ = 0.7505, C_5^+ = 0.5226.$	$a_3 > a_4 > a_5 > a_2 > a_1$
TODIM (Llamazares, 2018)	$\phi(a_1) = -3.9261, \phi(a_2) = -2.2095, \phi(a_3) = 1.3893,$ $\phi(a_4) = 0.0525, \phi(a_5) = -2.2994.$	$a_3 > a_4 > a_2 > a_5 > a_1$
VIKOR (Mishra et al., 2022)	$Q(a_1) = 1.0000, Q(a_2) = 0.7841, Q(a_3) = 0.0000,$ $Q(a_4) = 0.4284, Q(a_5) = 0.7880.$	$a_3 > a_4 > a_2 > a_5 > a_1$
MAGDM (Zha et al., 2021)	$Q(a_1) = 0.5548, Q(a_2) = 0.6483, Q(a_3) = 0.8129,$ $Q(a_4) = 0.7548, Q(a_5) = 0.6619.$	$a_3 > a_4 > a_5 > a_1 > a_2$
Quantum model (Wu et al., 2021a)	$P(a_1) = 0.1566, P(a_2) = 0.1847, P(a_3) = 0.2352,$ $P(a_4) = 0.2271, P(a_5) = 0.1963.$	$a_3 > a_4 > a_5 > a_2 > a_1$
Quantum model (Jiang and Liu, 2022)	$P(a_1) = 0.1535, P(a_2) = 0.1909, P(a_3) = 0.2407,$ $P(a_4) = 0.2241, P(a_5) = 0.1909.$	$a_3 > a_4 > a_5 = a_2 > a_1$
The proposed method in BN	$P(a_1) = 0.1564, P(a_2) = 0.1846, P(a_3) = 0.2374,$ $P(a_4) = 0.2262, P(a_5) = 0.1954.$	$a_3 > a_4 > a_5 > a_2 > a_1$
The proposed method	$P(a_1) = 0.1384, P(a_2) = 0.1909, P(a_3) = 0.1416,$ $P(a_4) = 0.3079, P(a_5) = 0.2212.$	$a_4 > a_5 > a_2 > a_3 > a_1$

Bayesian network, its ranking results are basically consistent with those of traditional decision making methods, which verifies the effectiveness and stability of the model proposed in this paper.

In contrast, the optimal choice for both the quantum model (Wu et al., 2021a) and the quantum model (Jiang and Liu, 2022) is a_3 , and there are significant differences between the ranking results and the alternatives in this paper. Specifically, the quantum model (Wu et al., 2021a) assumes that attributes' weights and decision-makers' weights are known, and also does not take into account possible changes in the mental state of decision-makers during the group consensus process. In addition, the interference effect measurement of the model on decision makers is also flawed, and the phase angle is not accurately measured, which leads to the decision accuracy and stability of the model in the face of complex decision situations. Therefore, although the quantum model (Wu et al., 2021a) can process uncertain information, it fails to fully reflect the interaction between decision makers and the synergistic effect in multi-attribute decision making, thus affecting the rationality of decision results. Although the quantum model (Jiang and Liu, 2022) takes into account the group consensus process, it fails to take into account the psychological state of decision makers. Especially in the face of group preference pressure, decision makers may have different psychological reactions, such as cognitive bias and group pressure, which may play a crucial role in the decision-making process. Secondly, the quantum model (Jiang and Liu, 2022) measures the interference angle between decision makers based on cosine similarity, but this method fails to accurately capture the nonlinear interference effect between decision makers. Cosine similarity usually assumes that the interference between decision makers is linear, which obviously does not fully fit the reality in complex decision situations, especially in high-dimensional, multi-attribute decision environments.

However, the method proposed in this paper considers the influence of psychological preference level on the consensus level of decision makers in complex uncertain decision-making situations. Secondly, based on the marginal contribution of each attribute, an optimization model is constructed to solve the optimal weights of the attributes. In addition, interference values between decision makers are measured based on Deng entropy. Therefore, the proposed method enhances the applicability and robustness of the model in complex decision scenarios, so as to be more consistent with the real complex situations.

5. Conclusion

In this paper, a multi-attribute quantum group consensus model is proposed to address the issues of information uncertainty and decision-maker interference during meteorological disaster emergency rescue. By introducing the linguistic Z-numbers, the model effectively handles uncertain information in complex decision-making environments. It

utilizes a group consensus model based on a minimum adjustment strategy, considering how the psychological state of decision makers under group preference pressure influences the feedback matrix. Decision makers' weights are determined by combining the distance between individual decision makers and the group preference matrix, as well as the marginal contribution of each decision maker. In addition, based on the marginal contribution of Shapley theory, an optimization model is constructed to solve the optimal attributes' weights, so as to optimize the rational distribution of attributes' weights under uncertain decision environment. Deng entropy is used to measure the interference term in quantum probability, ensuring accuracy in the decision-making process, while quantum probability is employed to rank alternatives, enhancing the scientific rigor of the decisions. The results demonstrate that the model not only manages uncertainty in emergency rescue situations but also promotes group consensus and improves decision-making efficiency through a reasonable adjustment mechanism.

This model has significant potential in various practical applications beyond meteorological disasters. The proposed multi-attribute quantum group consensus model can be adapted to other complex decision-making scenarios that involve high levels of uncertainty and diverse stakeholder opinions, such as supply chain management, disaster response strategies, policy-making, and collaborative business decisions. By accurately capturing decision-maker preferences, managing information uncertainty, and promoting consensus, this model can facilitate more efficient decision-making processes and improve outcomes in a wide range of domains. Moreover, its application in commercial decision-making environments, such as risk management and investment strategies, can provide businesses with a robust tool for optimizing decisions in uncertain and dynamic conditions. The model's ability to incorporate psychological preferences and address decision-maker interference further enhances its commercial value, offering a competitive edge in fields that require collaborative and complex decision-making processes.

While the multi-attribute quantum group consensus model proposed in this paper offers an innovative approach to handling uncertain information and decision interference, it has several limitations. (1) Quantum decision models often involve complex calculations and may require extensive computational resources, especially when dealing with a large number of decision-makers and attributes. The quantum superposition and entanglement phenomena used to model decision interference can significantly increase the computational complexity. (2) The psychological state modeling of decision makers in this study is relatively simplistic. It does not fully account for dynamic behavioral changes or nonlinear factors in complex decision-making environments, which may lead to an incomplete representation of decision makers' real responses under multi-dimensional pressures.

CRediT authorship contribution statement

Yizhao Xu: Writing – original draft, Visualization, Methodology.
Shuli Yan: Writing – review & editing, Validation, Supervision, Conceptualization.
Yangxue Li: Supervision, Conceptualization.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, and we do not have any possible conflicts of interest.

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Data availability

The authors do not have permission to share data.

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